CONSUMER PRICE INDEX (CPI) INFLATION ANALYSIS

Project Summary | 2017–2023

This project analyzes Consumer Price Index (CPI) inflation trends across different categories and time periods. The analysis focuses on overall inflation trends, food price movements, category-level CPI changes, the impact of COVID-19, and the relationship between fuel price fluctuations and inflation.

KEY OBJECTIVES

① Analyze the contribution of broader categories to overall CPI.

② Identify Year-over-Year (YoY) inflation trends.

③ Examine Month-over-Month (MoM) food inflation.

④ Measure changes in food subcategory CPI.

⑤ Compare Pre-COVID and Post-COVID inflation.

⑥ Analyze the relationship between Crude Oil price fluctuations and CPI categories and sub-categories using correlation analysis.

KEY FINDINGS

① Largest CPI Contributor	**Food – 51.73%**
② Highest YoY Inflation	**2022 – 6.62%**
③ Lowest/Latest YoY	**2023 – 3.18%**
④ Highest Food MoM Inflation	**Oct 2022 – 1.01%**
⑤ Lowest Food MoM Inflation	**Dec 2022 – -1.35%**
⑥ Largest Food CPI Increase	**Spices — +30.9 points**
⑦ Highest Category-Level Correlation	**Fuel and Light — 0.13**
Highest Sub-Category Correlation	**Meat and Fish — 0.39**

ANALYSIS COVERED

① **Category Contribution**	Food contributes 51.73% to the CPI basket.
② **YoY Inflation Trend**	Inflation peaked at 6.62% in 2022.
③ **Food MoM Analysis**	Food inflation showed significant monthly fluctuations.
④ **Food Sub-Category CPI Change**	Spices increased the most, while Oils & Fats declined significantly.
⑤ **COVID Impact**	Inflation increased across major categories after COVID-19.
⑥ **Fuel Correlation**	Meat and Fish showed the strongest positive relationship with Crude Oil price fluctuations among the analyzed sub-categories, with a correlation of 0.398.

CONCLUSION

The analysis shows that Food is the largest contributor to the CPI basket, while inflation peaked in 2022 and eased in 2023. Food categories experienced notable price fluctuations, and inflation increased after COVID-19. Correlation analysis indicates that Crude Oil price changes generally had a weak relationship with CPI movements, with Meat and Fish showing the strongest positive correlation (0.398).